# Attributing annotator polarization to sociodemographic groups

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## 1 Introduction

Annotations are essential for a wide range of tasks, especially in fields such as content moderation, toxicity detection and hate speech detection. These tasks are especially essential for training systems that protect vulnerable and minority groups in online spaces, where they are often targeted [17, 4]. However, annotations for such tasks are subjective, and are usually based on majority-group annotators. This may limit the usefulness of the data, or even lead to biased and misconfigured systems.

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree in vain. An example of such cases would be racist comments that are presented with coded language (usually referred to as "dog-whistles" [13]), which are not picked up by most annotators, but only by ones belonging in the targeted minority. Inversely, majority-group annotators may bias their own annotations against the minority-group annotators; CITATION for instance shows that toxicity models disproportionally marked African American speech as "toxic". This issue exists even in representative samples.

*Inter-annotator agreement* is often used to detect such cases [8]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [11, 12]—specifically whether two or more annotation clusters are present. Figure 1 displays how the two methods handle four common cases of annotation distributions for a hypothetical toxicity annotation task.

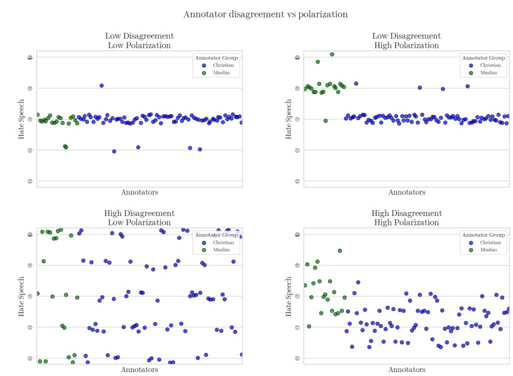


Figure 1: A comment annotated by 100 annotators. Disagreement is similar to variance; thus it overlooks cases where annotators may be "split down the middle". Polarization on the other hand, is able to identify multiple clusters, even if they are close together.

While we can detect polarization [12, 11], there is little work on how to detect whether it is actually caused by marginalized groups. Pavlopoulos and Likas [12] propose a mechanism which only works in the extreme case where overall polarization is zero; in this work we demonstrate that this case is not frequent in real-world data. We instead propose the *"Aposteriori Unimodality (Apunim) Metric"*, which attributes polarization to individual annotator sociodemographic groups. Our contributions are:

1. We introduce a new metric which measures how much polarization can be attributed specific sociodemographic groups.

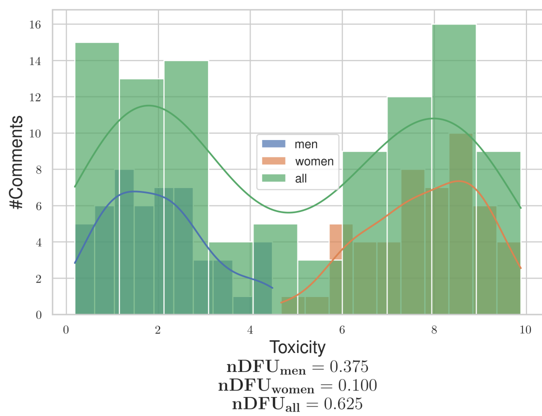


Figure 2: Example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

2. We discover issues not yet acknowledged in literature, such as the presence of inherent polarization and conflicting polarization directions. Our proposed metric is designed to avoid these issues.

3. We apply our metric to four datasets; two with human comments and annotations [15, 9] as well as two generated and annotated by Large Language Model (LLM) agents [16].

# 2 Methodology

In this section, we provide a formal mathematical formulation for the problem of attributing polarization to specific annotator characteristics (§2.1), and offer an intuitive rationale for how established polarization metrics can be leveraged (§2.2). We then explain how we can measure polarization for each group (§2.3) and how much of that polarization is caused by external variables (§2.4). Finally, we introduce our metric, which computes attribution on the entire dataset for each sociodemographic group (§2.5).

## 2.1 Problem Formulation

**SocioDemographic Backgrounds** Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to isolate individual groups within a SocioDemographic Background (SDB). Let $\Theta$ be one of multiple "dimensions" the set of all annotator SDB groups for a single dimension (e.g. "male", "female" and "non-binary" if we are investigating annotator gender).

**Annotations** Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated data points. We assume that annotating a data point depends on two variables: inherent ("apriori") controversy, and the ("aposteriori") annotator's SDB, as well as uncontrolled factors such as mood and personal experiences (noise). Assuming that each data-point $c$ is assigned multiple annotators, we can define the annotation multi-set $A(c) = \{(a, \theta)\}$, where $a$ is a single annotation. As an example, the annotations for a comment in a sentiment analysis task would be formulated as:

$$\begin{aligned} &A(\text{"}\textit{Could be better, could be worse}\text{"}) = \\ &\{(positive, female), (negative, male), \\ &(positive, female), \ldots\} \end{aligned} \tag{1}$$

**Polarization** Each set of annotations features a certain degree of *polarization*. Unpolarized annotations are usually unimodal; the annotators disagree on the details (which would be shown roughly as a bell curve around the median annotation), not fundamentally (which would likely be shown as a multimodal distribution). Pavlopoulos and Likas [12] create a polarization metric, the "normalized Distance From Unimodality (nDFU)", which measures whether an annotation set shows no polarization (unimodal distribution—$nDFU \approx 0$), up to complete polarization (multimodal distribution—$nDFU \approx 1$). Thus, we need a metric that measures how much the nDFU of a polarizing data point can be partially explained by $\theta$.

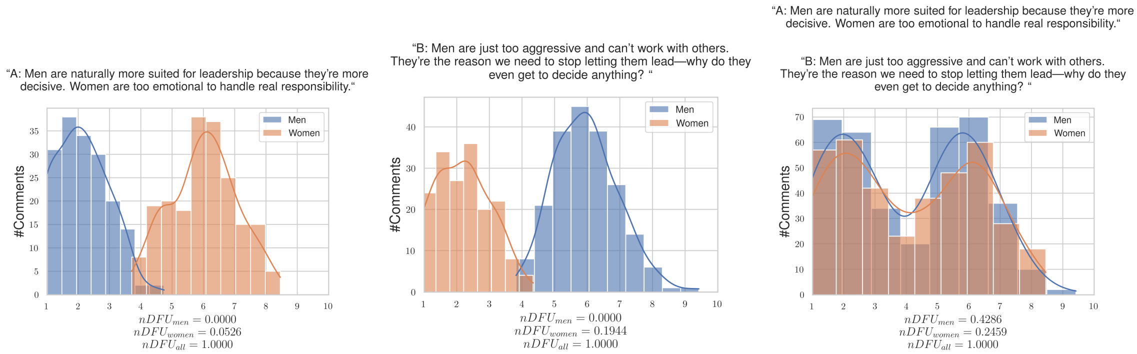


Figure 3: Hypothetical example of a polarizing discussion with two comments, for which the annotators disagree on which is the toxic one. If we aggregate the two comments, the polarization scores for both men and women significantly rise, obscuring whether the exhibited polarization can be partially attributed to gender.

## 2.2 Quantifying changes in polarization

**Comment polarization** Intuitively, $\theta$ partially explains the polarization of a data point $c$ when the annotations grouped by $\theta$ are more polarized compared to the full set of annotations. Figure 2 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators.[1] The annotations are generally polarized ($nDFU_{all} = 0.625$). The annotations by female annotators exhibit low polarization between themselves ($nDFU_{women} = 0.1$), since most agree the data-point is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.3725$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by disagreements between male and female annotators.

**Dataset polarization** Given this observation, we would be tempted to aggregate all annotations in the dataset and compare the polarization of each $\theta$ with the full set of annotations $A(c)$. In fact this is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each data point may be annotated by only 5 annotators, leading to a best-case 3-2 split between groups. Since nDFU is at its core a histogram estimation, 3 annotations may not be enough to acquire a reliable histogram (this is discussed in depth in §2.5).

However, this formulation would not work well. Figure 3 illustrates a hypothetical discussion with two comments, both of which are toxic, but where male and female annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This example demonstrates that polarization has a *direction*, which is not captured by the (symmetric) $nDFU$ metric.

## 2.3 Observed polarization

**Definition** The observed polarization of a datapoint on the group of annotations $\theta$ can be directly computed as:

$$pol(c, \theta) = nDFU(P(A(c), \theta)) \tag{2}$$

where $P(A(c), \theta_i) = \{a(c;\theta) \in A(c) | \theta = \theta_i\}$ is the partition of $A$ for a data-point $c$, for which its annotators belong to the SDB group $\theta_i$.

[1] The use of only two predominant genders is made only for the purposes of demonstration.

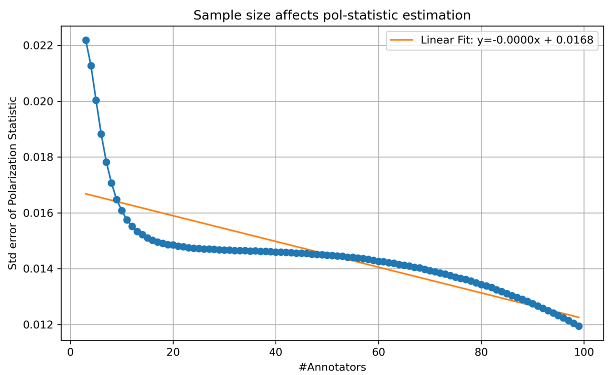


Figure 4: Standard error of the *pol-statistic* when sampled with replacement from the 100 LLM-annotator synthetic dataset for various number of annotators.

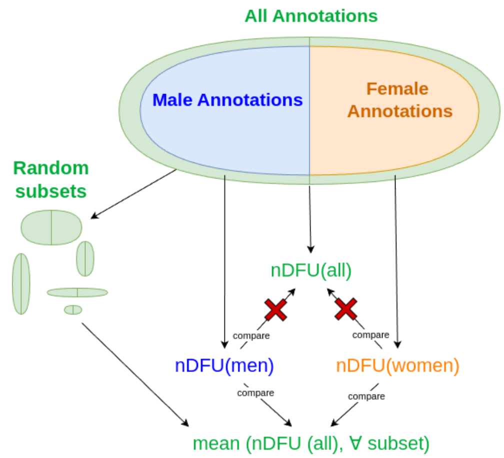


Figure 5: Since we can not compare the statistics of the group-subsets with the annotation superset, we elect to instead compare them with the mean apriori polarization of many sampled subsets.

**How many annotators?** While our test works best with many annotations per datapoint, the cost required for multiple human annotators is often prohibitive [14]. Figure 4 demonstrates the effect of the number of annotators on the reliability of the polarization estimation. We can see that the metric becomes consistent with about 15 annotators per comment, after which we start to get diminishing results. This experiment was conducted on a subset of the Sap et al. [15] dataset, with LLM annotators having different SDBs. The full experimental procedure can be found in Appendix D.

## 2.4 Apriori polarization

**Definition** In §1 we mentioned that annotator polarization can be caused either by the annotators SDB, or other factors. The latter can be considered "apriori" polarization, which is driven by the inherent "controversy" of the data point. Previous work [12] considered the case where this apriori polarization was zero; this is not the case in many datasets, as we demonstrate in §3.

**Statistical problems** In §2.2 we compared the polarization of the whole set of annotations with the partitions by annotator group. This approach uses the same samples for both subset and superset, violating the fundamental hypothesis of independent samples. This also means the statistics for both will *always be similar to an extent*. This is especially pronounced in our case, due to the extreme overlap exhibited by tiny annotation sets. Additionally, when calculating p-values, the non-independence of samples induces an intrinsic, non-zero covariance between any statistic $s$ calculated on $A$ and on $S$.

A standard method of side-stepping this issue is to compare the subset with its complement to the superset. This can not be applied to our case. Consider the example in Figure 2; if we compared $nDFU_{men}$ with $nDFU_{women}$, the result would be very close to zero, indicating no polarization attribution to gender.

**Overcoming dependent samples** Given infinite annotations, the assumption of independence would hold. Likewise, if we compared the subset with infinite random subsets of the superset. We can thus estimate the apriori polarization in data point $c$ by bootstrapping (Figure 5). We randomly partition the annotations in $|\Theta|$ groups, with matching group sizes (e.g., if we have 100 annotations, 80 of which are

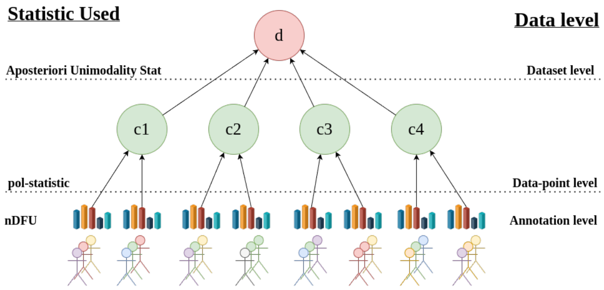


Figure 6: An overview of the Aposteriori Unimodality Test. We gather statistical information from the annotations (different SDBs are denoted by different colors) via the nDFU measure, which is aggregated on the data-point-level by the pol-statistic. The Aposteriori Unimodality Test is applied on the discussion level, operating on the pol-statistics of the individual data-points.

made by male annotators and 20 by female annotators, we will create random partitions of sizes 80 and 20). The randomly partitioned annotations are then sampled $t$ times. Formally, the estimated apriori polarization will be given by:

$$\frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{3}$$

where $\tilde{P}_i$ is the random partition operator with matching group sizes.

## 2.5 Aposteriori polarization

**Dataset filtering** While data scarcity is an important limitation when acquiring the observed polarization, this is decisively not the case with the number of data points themselves, which can range up to hundreds of thousands. Thus, we have the luxury of filtering data points before computing a dataset-wide metric. Since our metric attributes polarization, unpolarized data points are useless. The same applies for data points which are composed of only a single annotator group. Thus, we define the subset $S_d$ of dataset $d$ such that:

$$\begin{aligned} S_d = \{c \in d \mid nDFU(c) > \alpha \\ \textit{and } c \textit{ annotated by at least two groups}\} \end{aligned} \tag{4}$$

where $\alpha$ is a small positive number (in our experiments $\alpha = 0.2$).

**Definition** By obtaining the pol-statistics for all data-points in $S_d$ for SDB $\theta$, we can get the mean observed polarization for the dataset:

$$pol_{obs.}(d, \theta) = \frac{1}{|d|}\sum_{c \in S_d} pol(c, \theta) \tag{5}$$

and similarly the mean apriori dataset polarization:

$$pol_{apriori}(d) = \frac{1}{|d|}\sum_{c \in S_d} \frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{6}$$

We can then define the *"Aposteriori Unimodality" (Apunim)* Statistic as:

$$apunim(d, \theta) = \frac{pol_{obs.}(d, \theta) - pol_{apriori}(d)}{1 - pol_{apriori}(d)} \tag{7}$$

Since $nDFU \rightarrow [0, 1] \Rightarrow apunim \rightarrow [-1, 1]$. If $apunim(d, \theta) \approx 0$, the exhibited polarization among annotators of group $\theta$ does not surpass what would be expected by chance. If $apunim(d, \theta) > 0$, then the group partially explains a rise in polarization, and inversely for the case $apunim(d, \theta) < 0$. The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 6.

**p-value estimation** While it is possible to obtain p-value results for the apunim values being larger or less than 0 respectively, we find that the inherent variance of our statistic makes a p-value estimation worthless ($p \gg 0.1$). Nevertheless, we include a definition for parametric and non-parametric p-value estimation, and apply them to our datasets in Appendix B.

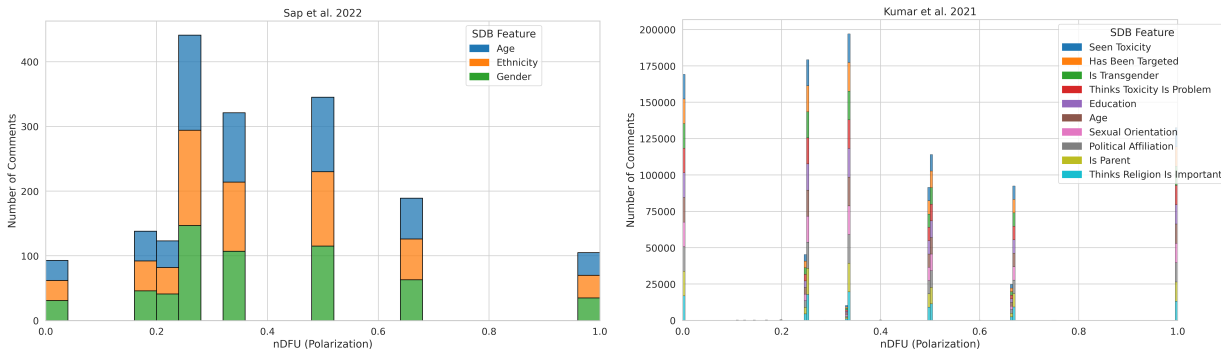


Figure 7: Histogram of dataset polarization for the human datasets considered in this paper per SDB feature.

# 3 Datasets

We use the datasets provided by Kumar et al. [9] and Sap et al. [15]. The datasets feature various online comments, each annotated by a number of human annotators (5 and 4-6 annotators per comment respectively). There is a reasonable amount of polarization in most annotations for both dataset, as demonstrated by Figure 7.

The Sap et al. [15] dataset includes racism annotations for 626 Twitter/X comments, and standard SDB information (age, race, education, gender). The Kumar et al. [9] dataset measures toxicity on various online comments, and the provided SDBs include mostly annotator experiences (e.g., whether they have been personally targeted online) as well as standard SDB information such as sexual orientation, age and education. Since the dataset is extensive (106035 comments), we only select a sample of 2000, due to computational constraints. Further analysis of the datasets can be found in Appendix TODO.

Table 1: Apunim results for the Sap et al. 2022 dataset

| | | apunim |
|---|---|---|
| SDB Feature | | |
| Age | 0-20 | — |
| | 20-40 | 0.0017 |
| | 40-60 | 0.0115 |
| | 60-80 | 0.0049 |
| Ethnicity | black | -0.2967 |
| | hisp | — |
| | middleEastern | — |
| | other | — |
| | white | 0.1400 |
| Gender | man | 0.0942 |
| | nonBinary | — |
| | woman | -0.1810 |

# 4 Results

Assuming a confidence level of $\alpha = 0.1$ we test whether any of the provided SDB dimensions can partially explain polarization in the toxicity/racism annotations. The results for the Kumar et al. [9] and Sap et al. [15] datasets can be found in Tables 1 and 2 respectively.

# 5 Conclusion

We introduced the “Aposteriori Unimodality Test”, a statistical test that detects whether certain sociodemographic groups partly cause polarization in annotations. Our test is resistant to low samples of annotations, bypasses common statistical issues such as sample dependence, and avoids newly-discovered issues such as the presence of inherent polarization and polarization direction. We apply our test two human-generated and two LLM-generated datasets and find interesting patterns on the former, and verify current literature on LLM SDB prompting on the latter.

# 6 Limitations

Since there are no other established tests that attribute polarization to sociodemographic characteristics, there is no way to verify that the results presented in §4 are accurate. Similarly, there is no qualitative way of evaluating our test, other than observations on the (non-) efficacy of LLM SDB prompting and our own intuition. Lastly, while our test seems to work sufficiently well with a small number of annotators per data-point, we still encourage researchers using this test to have at least a few annotators of each sociodemographic group that is being tested, in their dataset.

# 7 Ethical Considerations

While our works aims to help protect marginalized and disadvantaged groups by attributing polarization to certain subgroups, it can be taken advantage of by malicious actors. Given a dataset with fine-grained SDB information, these actors can use our test to target specific vulnerable groups. We thus urge researchers to keep datasets including such information protected, and provided to others only under explicit permission.

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# A Acronyms

**VMD** Virtual Moderation Dataset

**LLM** Large Language Model

**SDB** SocioDemographic Background

**nDFU** normalized Distance From Unimodality

**FWER** Family-Wise Error Rate

# B p-value Estimation

## B.1 Methodology

Like most metrics, we also include a p-value alongside the *apunim* value. This can be computed either non-parametrically, by repeatedly shuffling annotations and computing a null distribution of *apunim*, or parametrically, by assuming a normal distribution (which is generally a safe assumption if $t \geq 30$). In the latter case, we compute:

$$z = \frac{\hat{\kappa} - \mathbb{E}[\kappa]}{SE(\hat{\kappa})}, p = 1 - \Phi(z) \quad (8)$$

Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values. We choose the Holms method [6]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [3]). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

### B.2 Results

The full results for the datasets presented in §3, including p-value estimations, can be found in Tables 3 and 4 for the Sap et al. [15] and Kumar et al. [9] datasets respectively. The experiments were ran on the exact same parameters as in §4.

Note the absence of any statistically significant results for both parametric and non-parametric tests. This suggests that, although the differences in polarization captured by the *pol* statistic introduced in §2 are substantial enough to hint at polarization being attributable to certain groups, they remain too small to overcome the inherent noise in human annotations. As a result, we cannot obtain a robust estimate of statistical significance.

## C Synthetic Datasets

We use two synthetic datasets; one is the Virtual Moderation Dataset (VMD) presented in Tsirmpas, Androutsopoulos, and Pavlopoulos [16]. This dataset features 140 discussions, each having 14 comments (and usually up to 14 facilitator comments), each comment annotated by 10 LLM annotators, each supplied with a different SDB. The second dataset, is an extension of VMD, where we select four random unmoderated discussions, and employ 100 distinct LLM annotators.

An issue with synthetic datasets is the lack of overall polarization (see Figure 8). This can be offset by the ability of synthetic datasets to be scaled to a much greater extent than human datasets. This is because LLMs do not have cost and availability constraints of humans.

The results for the 100-annotator dataset can be found in Table **??**. We can not obtain results for the VMD dataset, since no comment exhibited apriori polarization while also being represented by more than two annotators of different groups. The lack of overall polarization, as well as the quantitatively smaller apunim values, suggest that SDB prompting can not be used to simulate annotations for human groups. This is consistent with relevant literature on the topic [1, 5, 14, 7, 2, 10].

## D LLM Annotation

TODO

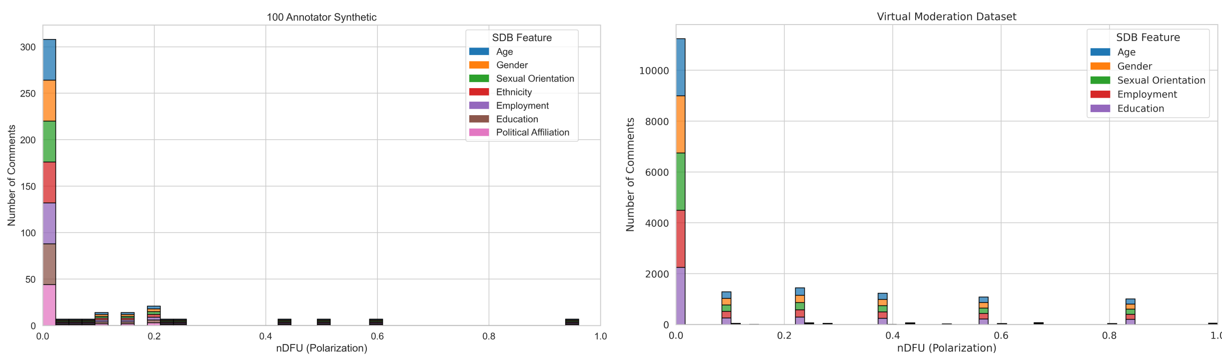


Figure 8: Histogram of dataset polarization for the synthetic datasets considered in this paper per SDB feature.

Table 2: Apunim results for the Kumar et al. 2021 dataset

| SDB Feature | | apunim |
|---|---|---|
| Age | 18 - 24 | -0.0081 |
| | 25 - 34 | 0.0220 |
| | 35 - 44 | -0.0176 |
| | 45 - 54 | 0.0079 |
| | 55 - 64 | -0.0077 |
| | 65 or older | -0.0022 |
| | Prefer not to say | — |
| | Under 18 | — |
| Education | Associate degree | -0.0066 |
| | Bachelor's degree | 0.0365 |
| | College, no degree | -0.0299 |
| | Doctoral degree | 0.0047 |
| | High School graduate | -0.0146 |
| | Master's degree | 0.0196 |
| | No high school | -0.0059 |
| | Other | — |
| | Prefer not to say | — |
| | Professional degree | -0.0001 |
| | NaN | — |
| Has Been Targeted | False | -0.0521 |
| | True | 0.0222 |
| Is Parent | No | -0.0677 |
| | Prefer not to say | 0.0051 |
| | Yes | 0.0513 |
| Is Transgender | No | -0.0703 |
| | Prefer not to say | 0.0036 |
| | Yes | -0.0027 |
| Political Affiliation | Conservative | 0.0287 |
| | Independent | -0.0159 |
| | Liberal | -0.0258 |
| | Other | -0.0085 |
| | Prefer not to say | 0.0040 |
| Seen Toxicity | False | 0.0386 |
| | True | -0.0751 |
| Sexual Orientation | Bisexual | 0.0101 |
| | Heterosexual | -0.0513 |
| | Homosexual | -0.0048 |
| | Other | 0.0006 |
| | Prefer not to say | -0.0053 |
| | NaN | 0.0093 |
| Thinks Religion Is Important | Not important | -0.0615 |
| | Not too important | -0.0126 |
| | Prefer not to say | 0.0005 |
| | Somewhat important | 0.0095 |
| | Very important | 0.0595 |
| Thinks Toxicity Is Problem | Frequently a problem | -0.0300 |
| | Not a problem | 0.0029 |
| | Occasionally a problem | -0.0033 |
| | Rarely a problem | 0.0142 |
| | Very frequently a problem | -0.0204 |

Table 3: Apunim results for the sap dataset

| SDB Feature | | apunim | p_param | p_nonparam |
|---|---|---|---|---|
| Age | 0-20 | — | — | — |
| | 20-40 | 0.0017 | 0.9958 | 0.9977 |
| | 40-60 | 0.0115 | 0.9794 | 0.9976 |
| | 60-80 | 0.0049 | 0.9660 | 1.0000 |
| Ethnicity | black | -0.2967 | 0.6369 | 0.6564 |
| | hisp | — | — | — |
| | middleEastern | — | — | — |
| | other | — | — | — |
| | white | 0.1400 | 0.7130 | 0.8246 |
| Gender | man | 0.0942 | 0.8368 | 0.8920 |
| | nonBinary | — | — | — |
| | woman | -0.1810 | 0.7198 | 0.7653 |

Table 4: Apunim results for the kumar dataset

| SDB Feature | | apunim | p_param | p_nonparam |
|---|---|---|---|---|
| Age | 18 - 24 | -0.0081 | 0.9808 | 1.0000 |
| | 25 - 34 | 0.0220 | 0.9696 | 0.9645 |
| | 35 - 44 | -0.0176 | 0.9719 | 0.9868 |
| | 45 - 54 | 0.0079 | 0.9819 | 1.0000 |
| | 55 - 64 | -0.0077 | 0.9765 | 1.0000 |
| | 65 or older | -0.0022 | 0.9917 | 1.0000 |
| | Prefer not to say | — | — | — |
| | Under 18 | — | — | — |
| Education | Associate degree | -0.0066 | 0.9847 | 1.0000 |
| | Bachelor's degree | 0.0365 | 0.9495 | 0.9334 |
| | College, no degree | -0.0299 | 0.9471 | 0.9960 |
| | Doctoral degree | 0.0047 | 0.9536 | 1.0000 |
| | High School graduate | -0.0146 | 0.9606 | 1.0000 |
| | Master's degree | 0.0196 | 0.9622 | 1.0000 |
| | No high school | -0.0059 | 0.9251 | 1.0000 |
| | Other | — | — | — |
| | Prefer not to say | — | — | — |
| | Professional degree | -0.0001 | 0.9994 | 1.0000 |
| | NaN | — | — | — |
| Has Been Targeted | False | -0.0521 | 0.9128 | 0.9751 |
| | True | 0.0222 | 0.9679 | 0.9851 |
| Is Parent | No | -0.0677 | 0.9050 | 0.9161 |
| | Prefer not to say | 0.0051 | 0.9698 | 1.0000 |
| | Yes | 0.0513 | 0.9256 | 0.9906 |
| Is Transgender | No | -0.0703 | 0.8548 | 0.9945 |
| | Prefer not to say | 0.0036 | 0.9776 | 1.0000 |
| | Yes | -0.0027 | 0.9897 | 1.0000 |
| Political Affiliation | Conservative | 0.0287 | 0.9552 | 0.9831 |
| | Independent | -0.0159 | 0.9755 | 0.9887 |
| | Liberal | -0.0258 | 0.9645 | 0.9489 |
| | Other | -0.0085 | 0.9650 | 1.0000 |
| | Prefer not to say | 0.0040 | 0.9821 | 1.0000 |
| Seen Toxicity | False | 0.0386 | 0.9382 | 0.9795 |
| | True | -0.0751 | 0.8667 | 0.9657 |
| Sexual Orientation | Bisexual | 0.0101 | 0.9777 | 0.9995 |
| | Heterosexual | -0.0513 | 0.9072 | 0.9883 |
| | Homosexual | -0.0048 | 0.9802 | 1.0000 |
| | Other | 0.0006 | 0.9917 | 1.0000 |
| | Prefer not to say | -0.0053 | 0.9771 | 1.0000 |
| | NaN | 0.0093 | 0.9065 | 1.0000 |
| Thinks Religion Is Important | Not important | -0.0615 | 0.9042 | 0.9446 |
| | Not too important | -0.0126 | 0.9706 | 1.0000 |
| | Prefer not to say | 0.0005 | 0.9960 | 1.0000 |
| | Somewhat important | 0.0095 | 0.9846 | 0.9945 |
| | Very important | 0.0595 | 0.9151 | 0.9563 |
| Thinks Toxicity Is Problem | Frequently a problem | -0.0300 | 0.9535 | 0.9730 |
| | Not a problem | 0.0029 | 0.9912 | 1.0000 |
| | Occasionally a problem | -0.0033 | 0.9953 | 1.0000 |
| | Rarely a problem | 0.0142 | 0.9735 | 1.0000 |
| | Very frequently a problem | -0.0204 | 0.9562 | 1.0000 |

Table 5: Apunim results for the 100 dataset

| SDB Feature | | apunim | p_param | p_nonparam |
|---|---|---|---|---|
| age | 0-20 | 0.0447 | 0.8422 | 0.8256 |
| | 20-40 | -0.0310 | 0.9492 | 0.9590 |
| | 40-60 | -0.0829 | 0.8636 | 0.8667 |
| | 60-80 | 0.0415 | 0.9300 | 0.9385 |
| education_level | bachelor's degree | -0.0649 | 0.8474 | 0.8564 |
| | doctoral degree | -0.0107 | 0.9614 | 0.9744 |
| | master's degree | -0.0155 | 0.9609 | 0.9538 |
| | no formal education | -0.0596 | 0.8557 | 0.8769 |
| | primary education | 0.0275 | 0.9392 | 0.9487 |
| | secondary education | 0.0197 | 0.9676 | 0.9846 |
| | vocational training | 0.0007 | 0.9970 | 1.0000 |
| gender | female | 0.0095 | 0.9833 | 1.0000 |
| | male | -0.0144 | 0.9807 | 0.9795 |
| | non-binary | 0.0298 | 0.8955 | 0.9231 |
| political_affiliation | apolitical | -0.0381 | 0.9238 | 0.9128 |
| | centrist | -0.0100 | 0.9851 | 0.9795 |
| | left-wing | -0.0205 | 0.9670 | 0.9641 |
| | right-wing | 0.0044 | 0.9899 | 0.9897 |
| sexual_orientation | asexual | 0.0031 | 0.9876 | 0.9949 |
| | bisexual | -0.0338 | 0.9017 | 0.9077 |
| | heterosexual | -0.0037 | 0.9973 | 0.9846 |
| | homosexual | 0.0319 | 0.9047 | 0.9231 |